

MATH 2630-080: TEST 2 – Summer 2023

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Name (*please print*): _____

Instructions: Please read the following instructions carefully.

- **Read all of the directions** for each problem.
- You have **75 minutes** to finish the exam.
- **Show work to receive credit.** Answers that are wrong, but still have work might receive partial credit. If you provide the answer to a computation without work, I will not give partial credit.
- Provide **exact** solutions.
- Do **not** write in the table below.

Problem	Points	Score
1	12	
2	12	
3	12	
4	10	
5	10	
6	10	
7	10	
8	12	
9	12	
Total	100	

1. (12 points) Find *all* first and second order partial derivatives of the function

$$f(x, y) = x^2 e^{xy}$$

2. (12 points) Find $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial u}$ at $s = 1, t = 2, u = 3$, if $z = x^4 + x^2y$, $x = s + 2t - u$, and $y = stu^2$.

3. (12 points) Find *all* local maximum, minimum, and saddle point values of the function

$$f(x, y) = x^2 - xy + y^2 + 9x - 6y + 5.$$

4. (10 point) Find the derivative of $f(x, y)$ in the direction of the vector $u = \langle 2, -4 \rangle$.

$$f(x, y) = 2x^3y - \frac{x}{y}$$

5. (10 points) Find the equation for the tangent plane and the normal line to the following surface at the point $(1, 1, 1)$.

$$yz + x \ln y = z^2.$$

6. (10) points) Evaluate the following limit. Write “*Does not exist*” if needed.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3 + x^3y}{x^4 + y^4 + x^2y^2}.$$

7. (10 points) Find f_{xyxy} if $f(x, y) = \sin(2x + 5y)$

8. (12 points) Consider the following 4-dimensional surface.

$$f(x, y, z) = 2x^3z + 3xyz + yz^2.$$

- (a) Find the gradient $\nabla f(x, y, z)$.
- (b) Find the direction of steepest ascent from the point $f(1, 1, -1)$.
- (c) Find the slope of the steepest ascent found in (b)

9. (12 points) Consider the following vector function.

$$\vec{r}(t) = \sin(2t)\hat{i} + \sqrt{12}t\hat{j} + \cos(2t)\hat{k}$$

- (a) Find the unit tangent, unit normal, and unit bi-normal vectors at the point $t = \pi$
- (b) Find the curvature of $\vec{r}(t)$ at $t = \pi$.